

Case Study: Core Banking System Modernization

Major Retail Banking Institution

CLIENT CONFIDENTIAL - BLINDED

Version 2.0 | Effective Date: December 2025

Meridian & Associates LLP | New York | Chicago | San Francisco | London | Singapore

This document is the proprietary and confidential property of Meridian & Associates LLP. It is intended solely for the use of the designated recipient(s). Any unauthorized reproduction, distribution, or disclosure of this material is strictly prohibited. The information contained herein is provided for informational purposes only and does not constitute a binding offer or agreement.

1. Engagement Overview

Client: Top-10 US retail bank (name withheld per confidentiality agreement)

Industry: Financial Services - Retail Banking

Engagement Type: Technology Transformation - Core Banking Modernization

Duration: 18 months (January 2024 - June 2025)

Team Size: 85 professionals at peak staffing

Total Fees: \$24.8 million

Lead Partner: Sarah Chen, CPA, CISA

Meridian Offices: New York (lead), Chicago, Hyderabad (GDC)

2. Client Situation & Challenge

The client, a top-10 US retail bank with \$285 billion in assets and 22 million consumer accounts, faced an urgent imperative to modernize its core banking infrastructure. The bank's transaction processing backbone had operated on an IBM zSeries mainframe platform for over 30 years, running custom COBOL applications that had been progressively layered and patched over three decades. While the system remained functionally stable, it presented escalating strategic and operational risks:

- **Technology Obsolescence:** The mainframe workforce was aging rapidly, with 40% of core COBOL developers expected to retire within 5 years. Recruiting replacements was increasingly difficult and expensive, with annual contractor rates exceeding \$280/hour for specialized mainframe skills.
- **Regulatory Pressure:** The OCC had issued a Matter Requiring Attention (MRA) citing the bank's inability to produce real-time transaction data for regulatory reporting. The existing batch-processing architecture introduced 24-48 hour latency in critical reporting feeds.
- **Customer Experience Gaps:** The bank's Net Promoter Score (NPS) had declined from 58 to 42 over three years, driven primarily by slow transaction processing, limited digital banking features, and an inability to offer real-time payments. Competitor banks with modern core systems were capturing market share in the 25-40 demographic.
- **Cost Structure:** Annual mainframe operating costs exceeded \$78 million, growing at 8-12% annually due to MIPS-based licensing models and specialized hardware maintenance. The bank estimated a 5-year total cost of ownership (TCO) of \$450 million to maintain the status quo.
- **Integration Constraints:** The monolithic mainframe architecture required custom point-to-point integrations for each new channel or partner, with average integration timelines of 6-9 months. This severely limited the bank's ability to partner with fintechs or launch new digital products.

3. Meridian's Solution

Meridian designed and executed a phased migration strategy to replace the legacy mainframe core with a cloud-native core banking platform while maintaining uninterrupted service to 22 million accounts. The solution architecture

comprised four integrated workstreams:

3.1 Core Banking Platform Selection & Implementation

After a rigorous 12-week evaluation of Temenos Transact, Thought Machine Vault, and Finacle, the team selected Thought Machine Vault as the target core banking platform. Key selection criteria included cloud-native architecture (Kubernetes-based), smart contract-driven product configuration, and real-time event-driven processing. Meridian led the platform implementation using an agile delivery methodology with 2-week sprints, deploying the platform on AWS EKS across three availability zones.

3.2 API Layer & Integration Architecture

Meridian designed and implemented a comprehensive API gateway layer using Kong Enterprise, exposing 450+ RESTful APIs for internal and external consumption. The API layer served as the integration backbone, decoupling the core banking platform from downstream channels (mobile, web, branch, ATM) and enabling fintech partner integration. An event-driven architecture using Apache Kafka processed over 2.8 million events per hour at peak, providing real-time data feeds for regulatory reporting, fraud detection, and customer analytics.

3.3 Data Migration & Parallel Run

The data migration workstream addressed 22 million customer accounts, 380 million historical transactions (7 years), and 14 product configurations. Meridian developed a custom migration framework using AWS DMS and custom Python ETL pipelines, executing three mock migrations before the production cutover. A 90-day parallel run period validated data integrity with 99.9997% accuracy before the legacy system was decommissioned.

3.4 Real-Time Payments Enablement

As part of the modernization, Meridian enabled the bank's connection to the FedNow instant payments network and upgraded the bank's RTP (Real-Time Payments) connectivity. The new architecture supported sub-second payment processing compared to the previous 2-4 hour batch cycle, positioning the bank as a leader in instant payments among top-10 US retail banks.

4. Delivery Approach & Team Composition

The engagement was structured as a multi-phase program with dedicated workstream leads, an integrated PMO, and a rigorous governance framework. Key delivery elements included:

- Agile delivery with 2-week sprints, PI planning every 10 weeks (SAFe-aligned)
- 85-person team at peak: 35 onshore, 50 offshore (Hyderabad GDC)
- Team composition: 2 Partners, 4 Senior Managers, 8 Managers, 22 Senior Consultants, 28 Consultants, 21 Analysts
- Weekly steering committee with C-suite sponsors (CTO, CIO, CFO)
- Bi-weekly regulatory updates to OCC relationship management team
- Independent quality assurance reviews at each phase gate

5. Technology Stack

The following technologies were deployed as part of the solution:

- **Core Banking:** Thought Machine Vault (cloud-native, smart contract-based)
- **Cloud Infrastructure:** AWS (EKS, RDS Aurora, S3, CloudFront, Route 53)
- **Containerization:** Docker, Kubernetes (EKS), Helm, ArgoCD
- **Event Streaming:** Apache Kafka (MSK), Schema Registry, Kafka Connect
- **API Management:** Kong Enterprise Gateway, OpenAPI 3.0 specifications
- **Data Migration:** AWS DMS, Custom Python ETL, Apache Spark
- **Monitoring:** Datadog (APM, infrastructure, logs), PagerDuty
- **Security:** HashiCorp Vault (secrets), AWS KMS, WAF, GuardDuty
- **CI/CD:** GitHub Enterprise, GitHub Actions, SonarQube, Artifactory

6. Results & Impact

The core banking modernization program delivered transformative results across all key performance dimensions, exceeding original business case projections in several areas:

Metric	Baseline (Pre)	Result (Post)	Improvement
Transaction Processing Time	4.2 seconds avg	2.5 seconds avg	40% reduction
Annual Infrastructure Cost	\$78M	\$46M	\$32M savings
System Uptime	99.92%	99.99%	Near-zero downtime
Customer NPS	42	67	+25 points
Regulatory Reporting Latency	24-48 hours	Real-time	Eliminated batch lag
API Integration Time	6-9 months	2-4 weeks	90% faster
Time to Market (new products)	9-12 months	4-6 weeks	85% faster

7. Client Testimonial

"Meridian's team brought an exceptional combination of deep banking domain expertise and cutting-edge technology skills. They didn't just migrate our core -- they fundamentally reimagined our technology architecture. The results speak for themselves: our operating costs are down, our customers are happier, and we're now launching new products in weeks instead of months. I would not hesitate to engage Meridian again for our next strategic initiative."

-- Chief Technology Officer, Client Institution